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Lammps_VSCODE: Interactive editor extension for LAMMPS molecular dynamics scripting

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Summary

Lammps_VSCODE is an extension package for Microsoft's popular code editor "Visual Studio Code" (VS Code) that supports materials scientists working with the molecular dynamics software LAMMPS. LAMMPS is an extensive, open-source molecular dynamics code with nearly 1000 high-level commands in its scripting environment. The code is constantly evolving with more than 250 contributors and regular release cycles. Using and understanding these commands in LAMMPS input scripts can be challenging, especially for beginners. Lammps_VSCODE provides an interface to LAMMPS and its extensive documentation and script syntax and implements functionalities such as syntax highlighting, tooltips, embedded offline documentation for all LAMMPS commands, autocomplete suggestions, some pre-run checks, and a simulation dashboard with interactive plotting of log data and 3D visualizations of atomic structure files. The extension is updated automatically through a continuous integration pipeline on a monthly schedule to ensure ongoing compatibility with the latest official LAMMPS release. Much of the syntax and functionality is also used in derivative packages such as LIGGGHTS (Kloss et al., 2012), where **Lammps_VSCODE** can also be used accordingly.

Statement of need

LAMMPS stands for "Large-scale Atomic/Molecular Massively Parallel Simulator" and is a classical molecular dynamics code. The development began in the 1990s at Sandia National Laboratories (Plimpton, 1995) and continues to this day as an open source project with a large number of contributors and an even larger number of users (Thompson et al., 2022). The scripting language used for setting up LAMMPS simulations has 1387 documented commands in the release branch of the GitHub repository at the time of writing and several parsing rules, which add a layer of complexity. The actual scripts are usually not particularly extensive and consist of four main parts:

1. Initializations
2. System definitions
3. Simulation settings
4. Simulation execution

However, in principle scripts can have arbitrary complexity and several features and conventions may not be obvious to new users. For example, compute commands or fixes with a given ID may be referenced by their ID with prefixes, such as `c_ID` and `v_ID` respectively, without declaring these variables explicitly. Variables of different types may be defined explicitly and referenced in the script. Expressions can also be evaluated directly without declaring variables, such as `$((xlo/xhi)/2+sqrt(v_area))`. In this example `$(...)` indicates a direct evaluation, `xlo`, `xhi` and `v_area` reference variables, `sqrt()` calls one of many functions known to LAMMPS, and mathematical operators are used and work as expected. Many more such rules may be

found in the documentation. These examples show that the LAMMPS scripting environment can indeed be seen as its own language, and syntax highlighting significantly helps navigating it. The commands implemented in LAMMPS may be easily used but often not intuitively understood. Quick access to the extensive documentation is therefore essential for productivity and learning. A direct interface for running and evaluating simulations within a familiar development environment supports this further. The advantages **Lammps_VSCODE** provides are appreciated by thousands of users and included in several online tutorials (Fahim, 2022; Mihok, 2022).

State of the field

Editor support for LAMMPS scripting has historically been limited. A few syntax highlighting packages for Sublime Text exist (BYU-BMR, 2017; Choudhary, 2017), but these have been unmaintained since 2017 and provide only basic keyword highlighting and snippets. An Emacs mode for LAMMPS, `lammps-mode` (Goswami, 2019), is available and provides syntax highlighting derived from `shell-script-mode`. One other VS Code extension for LAMMPS, `lammps-lang`, was published in 2019 but has been abandoned and offers only rudimentary syntax highlighting. None of these tools provide features beyond basic highlighting. **Lammps_VSCODE** is the only tool that offers comprehensive language support including autocompletion, embedded offline documentation, hover information, linting, a task provider for running simulations, and an interactive simulation dashboard with 3D visualization of atomic structures and live plotting of thermodynamic output.

Software design

Lammps_VSCODE is implemented in TypeScript using the VS Code Extension API. The extension is organized into modular components: a TextMate grammar file (`lmps.tmLanguage.json`) for syntax highlighting, providers for autocompletion, hover information, and diagnostics (linting), a documentation panel that renders LAMMPS command documentation as HTML within VS Code, and a simulation dashboard built with WebView panels for interactive plotting and 3D visualization.

A key design decision is the automated synchronization with the official LAMMPS documentation. A continuous integration pipeline runs monthly, parsing the reStructuredText documentation from the LAMMPS repository into structured JSON objects that drive the autocompletion, hover, and documentation features. This ensures that the extension stays current with the rapidly evolving LAMMPS codebase without requiring manual updates. The grammar file for syntax highlighting is also regenerated during this process from the latest command list. This approach was chosen over static keyword lists precisely because LAMMPS evolves continuously with frequent additions of new commands and styles.

Research impact statement

Since its initial release in 2019, **Lammps_VSCODE** has been installed over 47,000 times from the VS Code Marketplace. It has been recommended in several community tutorials on setting up LAMMPS workflows (Fahim, 2022; Mihok, 2022) and is used by researchers across materials science, chemistry, and physics. The extension lowers the barrier to entry for new LAMMPS users by providing an integrated development environment with immediate access to documentation, error checking, and simulation monitoring. Its broad adoption is evidenced by active community engagement through GitHub issues and feature requests from users worldwide.

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AI usage disclosure

The repository was designed and implemented over years by individuals without the use of AI tools. Some current commits, especially the creation of tests were assisted by GitHub Copilot using Anthropic's Claude Opus 4.5.

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